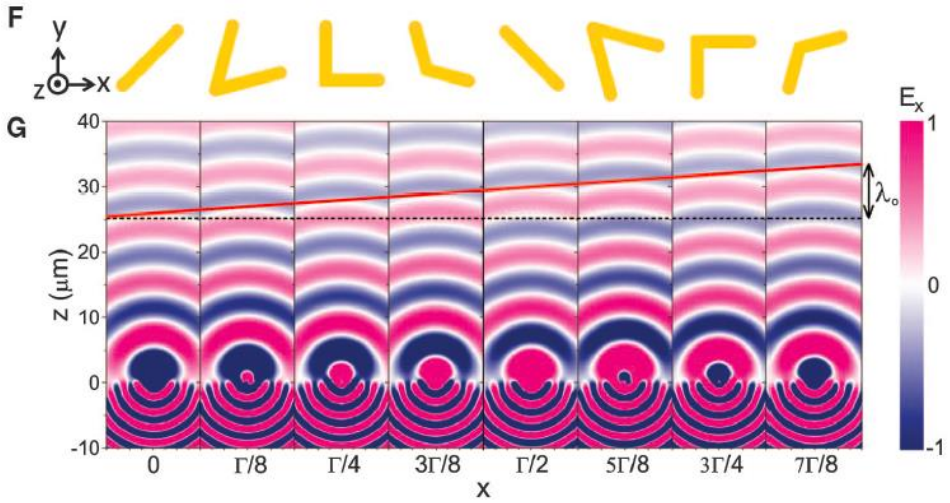
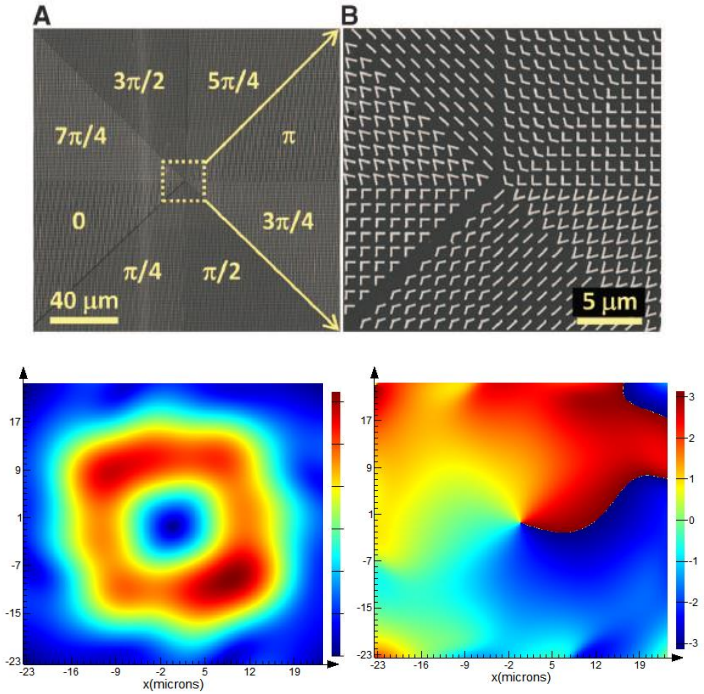
http://dx.doi.org/10.1126/science.1210713

涡旋光产生

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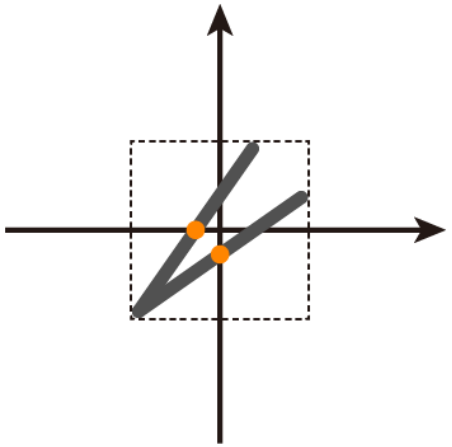




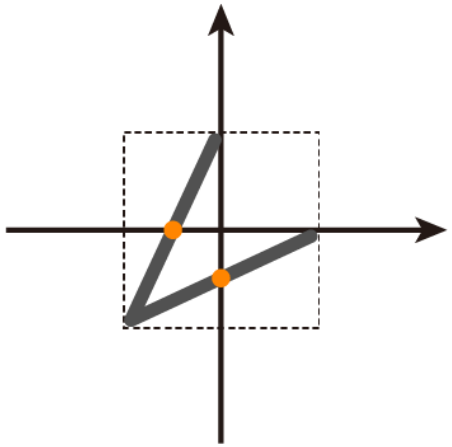
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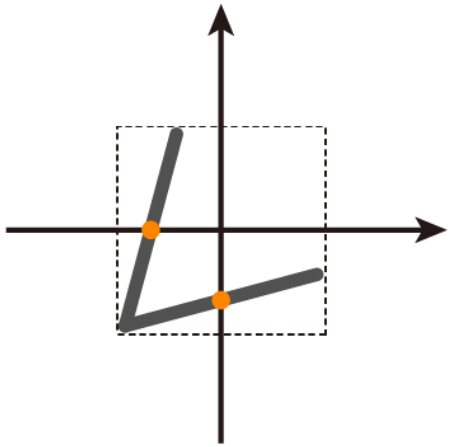
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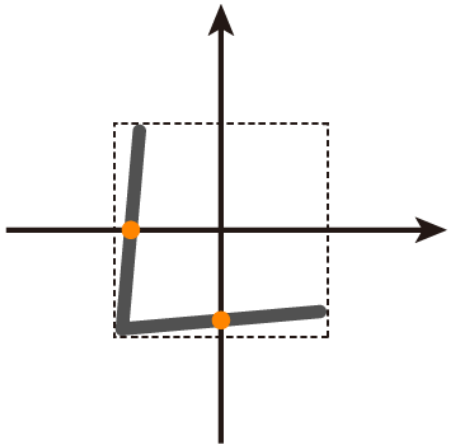
$\Delta = 20°$



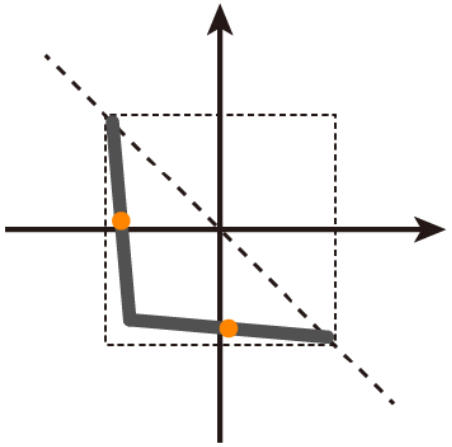
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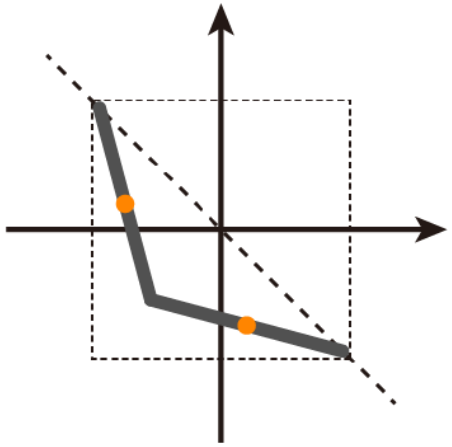
$\Delta = 60°$



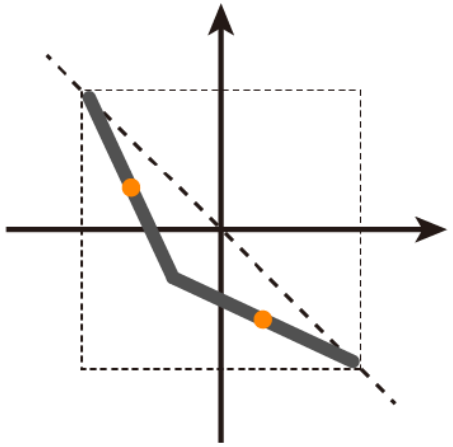
$\Delta = 80°$



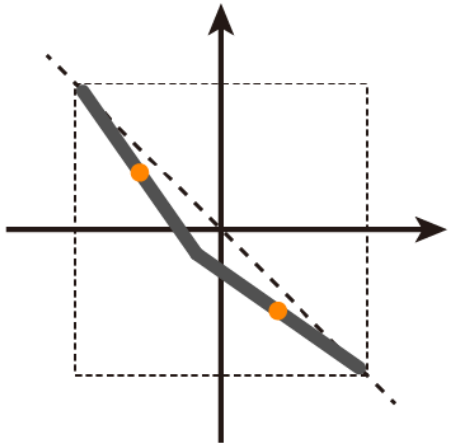
$\Delta = 100°$



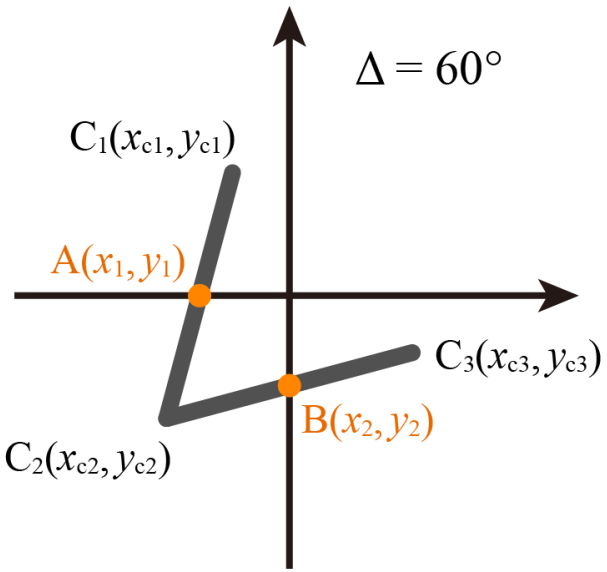
$\Delta = 120°$

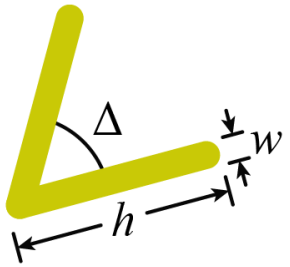


$\Delta = 140°$



$\Delta = 160°$





$$x_1 = -\frac{h-w}{2}\frac{\sin(\Delta/2)}{\sin(\pi/4)} \qquad y_1 = 0$$

$$x_2 = y_1 \qquad y_2 = x_1$$

$$x_{c1} = x_1 + \frac{h-w}{2}\sin(\frac{\pi}{4} - \Delta/2)$$

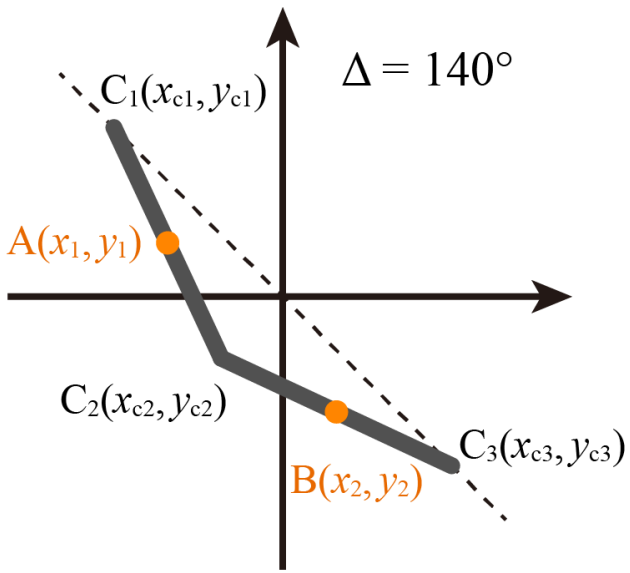
$$y_{c1} = y_1 + \frac{h-w}{2}\cos(\frac{\pi}{4} - \Delta/2)$$

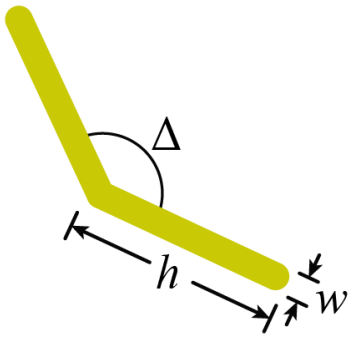
$$x_{c2} = x_2 - \frac{h-w}{2}\cos(\frac{\pi}{4} - \Delta/2)$$

$$y_{c2} = y_2 - \frac{h-w}{2}\sin(\frac{\pi}{4} - \Delta/2)$$

$$x_{c3} = x_2 + \frac{h-w}{2}\cos(\frac{\pi}{4} - \Delta/2)$$

$$y_{c3} = y_2 + \frac{h-w}{2}\sin(\frac{\pi}{4} - \Delta/2)$$





$$x_1 = -\frac{h-w}{2}\sin(\frac{3\pi}{4} - \Delta/2) \qquad y_1 = \frac{h-w}{2}\cos(\frac{3\pi}{4} - \Delta/2)$$

$$x_2 = y_1 \qquad y_2 = x_1$$

$$x_{c1} = x_1 - \frac{h-w}{2}\sin(\Delta/2 - \frac{\pi}{4})$$

$$y_{c1} = y_1 + \frac{h-w}{2}\cos(\Delta/2 - \frac{\pi}{4})$$

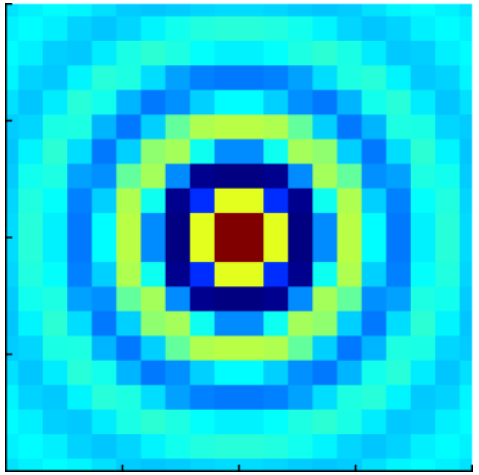
$$x_{c2} = x_2 - \frac{h-w}{2}\cos(\Delta/2 - \frac{\pi}{4})$$

$$y_{c2} = y_2 + \frac{h-w}{2}\sin(\Delta/2 - \frac{\pi}{4})$$

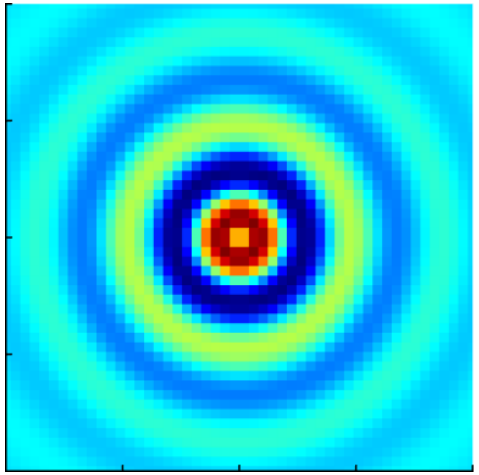
$$x_{c3} = x_2 + \frac{h-w}{2}\cos(\Delta/2 - \frac{\pi}{4})$$

$$y_{c3} = y_2 - \frac{h-w}{2}\sin(\Delta/2 - \frac{\pi}{4})$$

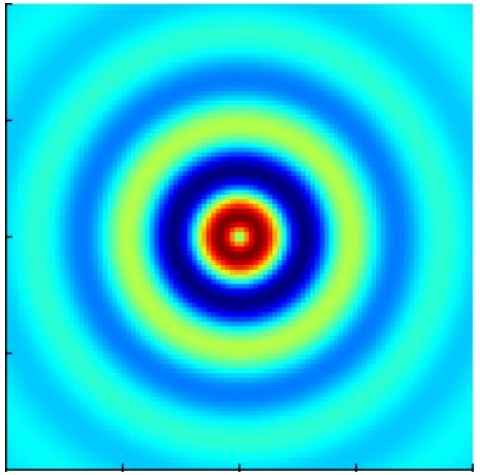
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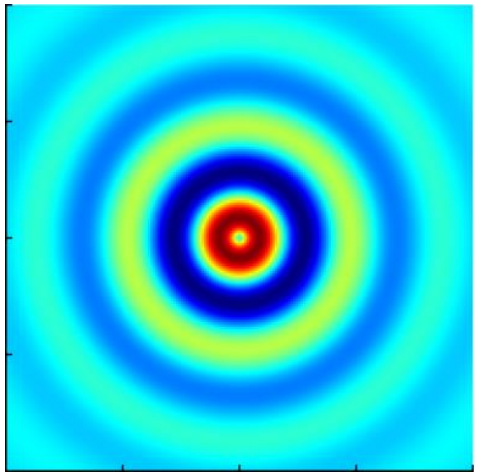
20×20



50×50

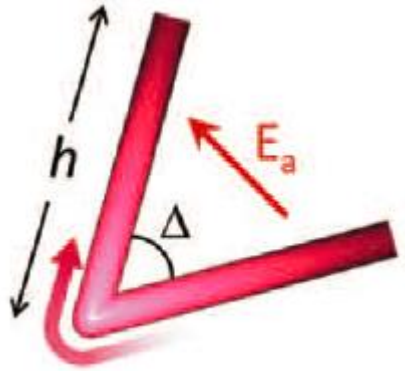


100×100



200×200

每个 V 型结构的棒长和夹角

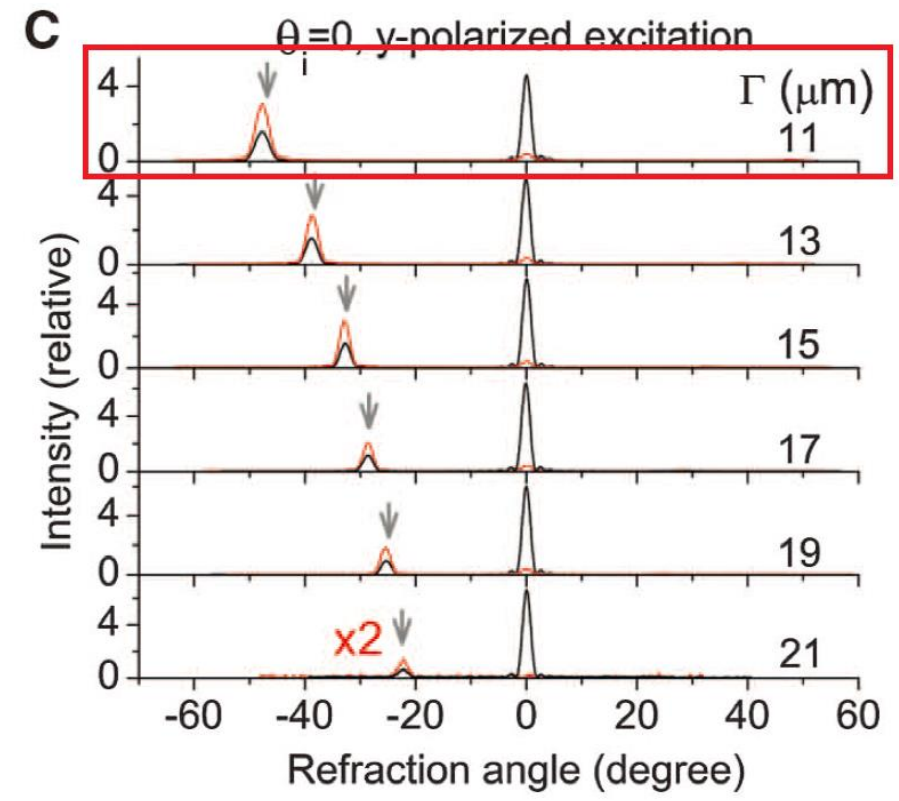




h/nm	760	1350	1130	940	760	1350	1130	940
Δ/deg	180	60	90	120	180	60	90	120

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Fig.3 的仿真方法见第七讲



The red and black curves are measured with and without a polarizer, respectively

用第七讲中的远场方法得到的结果：

